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RBIF114 – Final

Introduction

The TCGA-PRAD dataset, part of The Cancer Genome Atlas, provides a comprehensive molecular profile of prostate adenocarcinoma, a common malignancy affecting approximately one in nine men. Most diagnoses occur after age 66, and the disease disproportionately affects African American men [1]. Advances in screening and molecular profiling have improved early detection and risk stratification, increasing interest in identifying biomarkers that can distinguish tumor from normal tissue and help inform diagnosis, prognosis, and treatment decisions [1]. In this analysis , RNA-seq data from the TCGA-PRAD dataset were analyzed to identify genes and pathways that differentiate prostate tumor tissue from matched normal samples. The analysis integrates several complementary approaches, including differential expression analysis, pathway enrichment, gene set enrichment analysis (GSEA), and gene co-expression network analysis using WGCNA. This workflow moves from identifying individual differentially expressed genes, to interpreting pathway-level biological changes, and finally linking gene expression patterns to clinical traits through co-expression modules. One strength of this study design is the use of patient-matched tumor and normal samples, which helps reduce inter-patient variability and improves the biological relevance of the comparisons. However, restricting the analysis to matched samples reduces the dataset to approximately 52 pairs, which may limit statistical power in some enrichment analyses, such as KEGG over-representation tests. In this context, correlation and network-based approaches, such as WGCNA, can be particularly useful for identifying coordinated transcriptional programs that may not be detected through single-gene analyses alone.

Data Analysis:

The TCGA-PRAD dataset includes 501 primary tumor samples, 52 solid tissue normal samples, and one metastatic sample. To enable a controlled comparison, patients with both primary tumor (TP) and matched normal tissue (NT) samples were identified using the first 12 characters of TCGA barcodes, resulting in approximately 52 matched tumor–normal pairs. The dataset was then filtered to retain protein-coding genes, reducing noise from low-confidence or noncoding features.

Differential expression analysis was performed using a limma-based pipeline integrating Bioconductor packages including edgeR, limma, and SummarizedExperiment. Count data were normalized using the trimmed mean of M-values (TMM) method to correct for differences in library size and composition across samples. The normalized counts were then transformed using the voom method, which estimates the mean–variance relationship and generates $\log_2$-transformed expression values with precision weights suitable for linear modeling.The top 100 differentially expressed genes were extracted and ranked by p-value, representing the most statistically significant expression differences between tumor and normal tissues.

<u>Fold-Change Analysis:</u>

Fold-change filtering of the top 100 differentially expressed genes using FDR < 0.05 and $|\log_2 FC| > 1$ retained 88 genes with strong statistical and expression differences. DLX1, ZIC2, and NKX2–3 showed the largest positive fold changes, while CRYAB, ID4, and AOX1 showed the strongest negative fold changes in tumor.

Clinically, DLX1, ZIC2, and NKX2–3 have been reported to be upregulated in prostate cancer and are associated with oncogenic signaling and poorer patient outcomes, including activation of the Wnt/β-catenin pathway and potential response to anti–PD-1 immunotherapy [2,4,5]. In contrast, CRYAB, ID4, and AOX1 appear to function as tumor suppressors, with reduced expression linked to metastasis, increased tumor progression, and poorer survival outcomes in prostate cancer patients [3,6,7].

Principal Component Analysis and Volcano Plot:

Principal component analysis (PCA) was performed using the voom-normalized expression matrix generated from the limma pipeline. The PCA plot (PC1 vs. PC2) shows clear separation between primary tumor samples (black) and matched normal tissues (pink), indicating distinct transcriptional profiles of the different sample types (Figure 1).

A volcano plot illustrates differential gene expression between tumor and normal samples (Figure 2), with upregulated genes shown in red and downregulated genes in blue. The clustering at high −log10(adjusted p-values) indicates a large number of significant genes.

Hierarchical Clustering

Hierarchical clustering was performed on the 88 genes that met both the FDR (<0.05) and fold-change ($|\log_2 FC| > 1$) thresholds (Figure 3). Genes were ranked by decreasing $\log_2$ fold change, and their expression values across all samples were used for clustering based on Euclidean distance.. The dendrogram reveals several major branches, suggesting that these genes form broader expression modules. This pattern indicates that tumor-associated transcriptional changes occur through coordinated shifts in groups of genes, likely reflecting shared biological processes such as cell proliferation, metabolism, and loss of normal tissue function.

Over Representation Analaysis with Function annotation (GO & KEGG)

The filtered gene set was divided into upregulated and downregulated groups based on fold-change direction. Over-representation analysis was conducted separately for each gene group across all GO categories, including Biological Process, Molecular Function, and Cellular Component, with Benjamini–Hochberg correction applied to control for multiple testing. The most significant enriched terms were visualized using dot plots displaying the top ten GO categories for each group (Figure 4). The enrichment analysis suggests that tumor-associated DEGs are strongly enriched in developmental and differentiation-related programs. GO terms for upregulated genes include pattern specification, regionalization, and neuronal or glial differentiation, indicating that tumor cells may be reactivating embryonic lineage programs (figure 4.a). For example, pattern specification, which showed the highest gene count in the GO enrichment results, refers to developmental processes that establish spatial identity and guide cell differentiation [8]. This observation is consistent with evidence that cancer cells can reactivate embryonic signaling pathways, such as Wnt signaling, that are normally inactive in differentiated tissues [9]. In prostate cancer, loss or epigenetic silencing of the tumor suppressor APC disrupts β-catenin degradation, while activating mutations in β-catenin further promote early tumor development [9]. These results suggest that tumors may adopt developmental or stem-like transcriptional states. In this dataset, GO enrichment appears more statistically robust than KEGG enrichment, as indicated by consistently lower q-values. KEGG enrichment for upregulated genes was weaker, with higher adjusted p-values and smaller gene counts, suggesting that the tumor transcriptome is more strongly characterized by developmental and cell identity programs rather than metabolic pathways (figure 5.a). These findings align with the fold-change analysis, which highlighted upregulated genes involved in developmental and oncogenic signaling pathways.

For the downregulated genes, GO enrichment highlights processes related to membrane organization and vesicular trafficking, including receptor-mediated endocytosis and membrane raft assembly (Figure 4.b). This suggests altered receptor internalization and disrupted membrane dynamics in tumor tissue compared with normal prostate tissue. Supporting this observation, Boyu et al. reported increased clathrin expression and reduced caveolin-1 (Cav-1) levels in prostate cancer, where enhanced clathrin-mediated EGFR recycling was associated with increased tumor aggressiveness [10]. Additionally, KEGG

enrichment indicated reduced activity in detoxification and metabolic pathways, including cytochrome P450 metabolism, xenobiotic metabolism, and glutathione metabolism (Figure 5.b). Collectively, these results suggest that tumors may exhibit both altered receptor signaling regulation and diminished normal metabolic capacity relative to normal tissue.

Gene Set Enrichment Analysis (GSEA):

The GSEA results provide a broader, pathway-level view of how tumor samples differ from normal tissue. In the GSEA GO analysis, the strongest signals are centered around mitosis and chromosome segregation, which is consistent with increased proliferation and altered chromosomal regulation as key features of the tumor transcriptome (Figure 6.a). It overall points to coordinated activation of cell cycle programs across many genes. The KEGG analysis supports a similar theme but at a more pathway-oriented level. The enriched pathways include cell cycle, pathways in cancer, Calcium signaling, and Ribosome, may reflect on the enhanced growth signaling, structural remodeling, and increased protein synthesis associated with a highly proliferative and transcriptionally rewired tumor state (Figure 6.b). In result, the GSEA summarizes that tumors are driven by coordinated activation of proliferation and cancer-related pathways, reflecting consistent biological shifts rather than random expression changes.

Gene Correlation Network and STRING Comparison

A gene–gene correlation network was constructed using Pearson correlation to investigate relationships amongst the significant genes. An adjacency matrix was generated using a correlation threshold of 0.7 with self-correlations removed. In this network, nodes represent genes and edges represent co-expression relationships. For exploratory analysis, the network was reconstructed in igraph, where network statistics were calculated for each gene, including in-degree, out-degree, and betweenness centrality. Degree measures the number of connections a gene has within the network, while betweenness reflects how important a gene is for linking different regions of the network. In result, genes were then ranked by in-degree and out-degree to identify the most connected hub genes (Table 1).

Genes with high in-degree receive many connections and often represent convergence points of shared co-expression patterns, acting as downstream effectors or markers of coordinated biological programs. In this network, APOBEC3C emerged as the top incoming hub. Pang et al. reported that APOBEC3C is downregulated in prostate cancer and that reduced expression is associated with advanced tumor stage, higher Gleason scores, and poorer survival [11]. APOBEC3C has been linked to an activated anti-tumor immune microenvironment, including increased $CD8^+$ T cell infiltration and reduced M2 macrophages [11]. It also furthers activates downstream signaling pathways involved in DNA damage response and inflammatory signaling, highlighting important characteristics of a hub gene receiving a lot of incoming influences of the network. On the otherhand, genes with high out-degree connect to many other genes and may act as regulatory or signaling hubs that influence multiple downstream pathways. In this network, GSTP1 ranked as the top outgoing hub. GSTP1 functions in detoxification and interacts with signaling pathways that regulate proliferation, differentiation, and apoptosis. Filippo et al. demonstrated that GSTP1 is frequently hypermethylated in prostate cancer, leading to reduced expression compared with normal prostate tissue [12]. Because also GSTP1 belongs to the GSTs family, a group of enzymes, is consistent with its high out-degree in the network.

the correlation-based gene network was compared with a known protein–protein interaction (PPI) network from the STRING database to determine whether strongly co-expressed gene pairs were supported by known biological interactions. A STRING database object for human genes was created, and genes from the correlation network were used as the input set. These genes were mapped to

STRING protein identifiers to retrieve known interactions. To ensure reliability, STRING interactions were filtered to include only high-confidence edges with a combined score ≥700.

To compare the networks, both the correlation edges and STRING interactions were converted into undirected gene-pair identifiers, allowing overlapping interactions to be identified regardless of gene order. The correlation network contained 2010 edges, while the STRING PPI network contained six high-confidence interactions among the same gene set. Of these, three gene pairs overlapped: GSTM4–GSTP1, CRYAB–HSPB8, and ANXA2–S100A6. These overlapping pairs indicate genes that are not only strongly co-expressed in the dataset but also supported by known protein-level interactions.

Although the overlap between the networks was relatively small, this is expected because correlation networks capture co-expression relationships, which may reflect shared regulation, pathway membership, or indirect functional associations. In contrast, the STRING database focuses on experimentally supported or predicted physical protein interactions, making it more conservative.

Functional annotations of the overlapping pairs provide biological context for these relationships. The results from STRING database showed that GSTM4 and GSTP1 belong to the glutathione S-transferase family and are involved in detoxification and oxidative stress responses. CRYAB and HSPB8 are small heat shock proteins with chaperone activity, functioning in protein quality control and cellular stress responses. ANXA2 and S100A6 participate in calcium-dependent signaling and cytoskeletal regulation, processes that are frequently altered during tumor progression. In all, these interactions highlight pathways associated with cellular stress responses, metabolism, and tumor-related signaling, supporting the biological relevance of the co-expression network.

WGCNA Network Construction and Selected Module Analysis

A soft-thresholding power of 20 was selected for WGCNA network construction based on the scale-free topology analysis (Figure 7). At this power, the scale-free topology fit (R^2) approached the target threshold, indicating that the network approximates a scale-free structure, while mean connectivity remained sufficient to preserve meaningful gene–gene relationships. This balance supports the use of power 20 for downstream co-expression network construction and module detection. The WGCNA dendrogram identifies several gene co-expression modules (Figure 8), including prominent blue and yellow modules along with smaller red and turquoise clusters, while many genes remain unassigned in the grey module.

The heatmap shows the correlation between WGCNA module eigengenes and clinical traits, including tumor status, age, Gleason score, and race (Figure 9). Several modules show strong associations with tumor status. MEgreen, MEgrey, and MEturquoise are strongly negatively correlated with tumor samples, while MEpink and MEbrown show strong positive correlations with tumor presence, suggesting that these modules represent tumor-associated gene expression programs. Notably, MEpink also shows a moderate positive correlation with Gleason score ($r = 0.33$), indicating a potential relationship with tumor aggressiveness. In contrast, age and race show relatively weak correlations with most modules, suggesting they have limited influence on the detected co-expression patterns.

The MEpink module genes were further evaluated based on their association with tumor status and Gleason score. The gene significance values were calculated and combined into a table containing gene names, module membership, and correlation values. The genes were then ranked according to their combined association with tumor presence and disease severity. A scatter plot was generated to

visualize the relationship between gene significance for tumor status (x-axis) and gene significance for Gleason score (y-axis), with the top genes labeled (Figure 10).

Several genes within the MEpink module, including HJURP, MKI67, RRM2, NEIL3, MYBL2, and CDC25C, showed strong associations with both tumor status and Gleason score. Among these, HJURP (Holliday Junction Recognition Protein) stood out as a key candidate. Previous studies have reported that HJURP is upregulated in prostate cancer, and higher expression levels are associated with poorer patient outcomes [13]. Functionally, HJURP promotes tumor cell proliferation by enhancing degradation of the cell-cycle inhibitor CDKN1A (p21) through the GSK3β/JNK signaling pathway, supporting its potential role in tumor progression [13].

MEpink module was also shown to be more active in higher-grade tumors when module activity was examined across Gleason scores. The trend observed that MEpink activity increases with tumor grade, with the highest activity observed in Gleason score 9 tumors (0.115), followed by Gleason 8 (0.014), while lower scores such as Gleason 7 (-0.009) and Gleason 6 (-0.032) show reduced activity. This pattern suggests that the MEpink module becomes progressively more active as tumors become more aggressive, further supporting its potential role in prostate cancer progression.

Conclusion:

This analysis used a multi-layered transcriptomic workflow to examine gene expression differences between patient-matched tumor and normal prostate tissues using TCGA-PRAD RNA-seq data. Differential expression analysis identified several genes with strong statistical and biological relevance, including DLX1, ZIC2, and NKX2–3 as the most strongly upregulated and CRYAB, ID4, and AOX1 as the most downregulated in tumor samples. Elevated expression of DLX1, ZIC2, and NKX2–3 has been associated with oncogenic signaling and poorer clinical outcomes, while reduced expression of CRYAB, ID4, and AOX1 reflects loss of tumor-suppressive activity and is linked to more advanced disease states. Pathway enrichment analyses further showed that upregulated genes were enriched in developmental and proliferative pathways, whereas downregulated genes were associated with membrane organization and metabolic regulation. These expression patterns supports DLX1, ZIC2, NKX2–3, CRYAB, ID4, and AOX1 as promising candidates for prostate cancer biomarker discovery and disease progression monitoring.

Next, the network-based analyses provided additional insight into the organization of tumor-associated transcriptional programs. The gene–gene correlation network identified several highly connected hub genes, including APOBEC3C and GSTP1, which may help coordinate transcriptional responses in the tumor environment. Furthermore, the comparison with known protein–protein interactions from the STRING database revealed overlapping interactions associated with oxidative stress response, protein quality control, and calcium-dependent signaling, supporting the biological relevance of the co-expression network. Weighted Gene Co-expression Network Analysis (WGCNA) further identified modules (MEpink) associated with tumor status and disease severity. The resulting MEpink module showed a strong positive association with tumor presence and a moderate correlation with Gleason score, suggesting a link with tumor aggressiveness. Within this module, HJURP emerged as a key gene, consistent with previous studies reporting its role in promoting tumor cell proliferation and its association with poorer clinical outcomes in prostate cancer.

While many of the genes identified in this analysis have already been linked to prostate cancer biology, combining differential expression analysis with pathway enrichment, co-expression networks, and module–trait relationships provides a broader view of how these genes function within coordinated

transcriptional programs. This systems-level perspective highlights potential biomarkers and regulatory hubs, such as DLX1, GSTP1, and HJURP, that may have diagnostic or prognostic value if their expression correlates with disease severity or treatment response.

One limitation of this analysis relates to the WGCNA network construction. The dendrogram suggests a relatively sparse network that may be limited by the selected high soft-thresholding power (20) used to approximate scale-free topology. The analysis can be further optimized by exploring with lower soft thresholds powers such as 10, to produce a more connected network and to potentially identify more co-expression modules. Another interesting direction would be to further examine the MEblack module, which showed a moderate correlation with the Black/African American clinical trait ($r = 0.25$). Because prostate cancer disproportionately affects African American men, investigating genes within this module may help identify transcriptional patterns linked to population-specific differences in disease biology [1].

Future studies could further validate and expand these findings by testing the gene signatures in independent patient cohorts and experimentally confirming expression changes using qPCR, Western blotting, or immunohistochemistry. Functional experiments, such as CRISPR or siRNA knockdown, could also help clarify the roles of key genes like HJURP or DLX1 in prostate cancer progression.

Material and methods:

Data acquisition and preprocessing

RNA-seq data for prostate adenocarcinoma were obtained from the TCGA-PRAD dataset using gene-level counts generated by the STAR pipeline. Metadata indicated 501 primary tumor samples, 52 solid tissue normal samples, and one metastatic sample. Patients with both primary tumor (TP) and matched normal tissue (NT) samples were identified using the first 12 characters of TCGA barcodes, resulting in approximately 52 matched tumor–normal pairs. The dataset was filtered to retain protein-coding genes.

Differential expression analysis

Differential expression was performed using a limma-based pipeline integrating edgeR, limma, and SummarizedExperiment. Lowly expressed genes were filtered using filterByExpr(), and counts were normalized using TMM normalization. Data were transformed using voom to generate weighted $\log_2$ expression values for linear modeling. Gene-wise linear models were fitted with lmFit and moderated using eBayes. Differentially expressed genes were ranked using topTable, producing a full ranked list and the top 100 significant genes.

Fold-change filtering and visualization

Genes were further filtered using FDR < 0.05 and $|\log_2 FC| > 1$, resulting in 88 genes with strong statistical and fold-change support. PCA and volcano plots were used to visualize transcriptional differences between tumor and normal tissues. Hierarchical clustering with Euclidean distance was performed to identify groups of genes with similar expression patterns.

Functional enrichment and pathway analysis

Genes were separated into upregulated and downregulated groups and analyzed using GO and KEGG enrichment. Gene symbols were mapped to Entrez IDs using org.Hs.eg.db, and enrichment analysis was performed using enrichGO() with Benjamini–Hochberg correction. Gene Set Enrichment Analysis (GSEA) was also performed using the full ranked gene list to identify pathway-level differences.

Network analysis

A gene–gene correlation network was constructed using Pearson correlation ($r \geq 0.7$) to identify hub

genes. Network metrics were calculated using igraph, and results were compared with high-confidence protein–protein interactions from the STRING database.

Weighted Gene Co-expression Network Analysis (WGCNA)

The voom-normalized expression matrix was used to construct a co-expression network with WGCNA. Genes were filtered by median absolute deviation (MAD), and pickSoftThreshold() was used to select a soft-thresholding power (β = 20) based on scale-free topology. Modules were identified using blockwiseModules(), and module eigengenes were correlated with clinical traits including tumor status, Gleason score, age, and race to identify disease-associated gene modules.

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Figure 1:

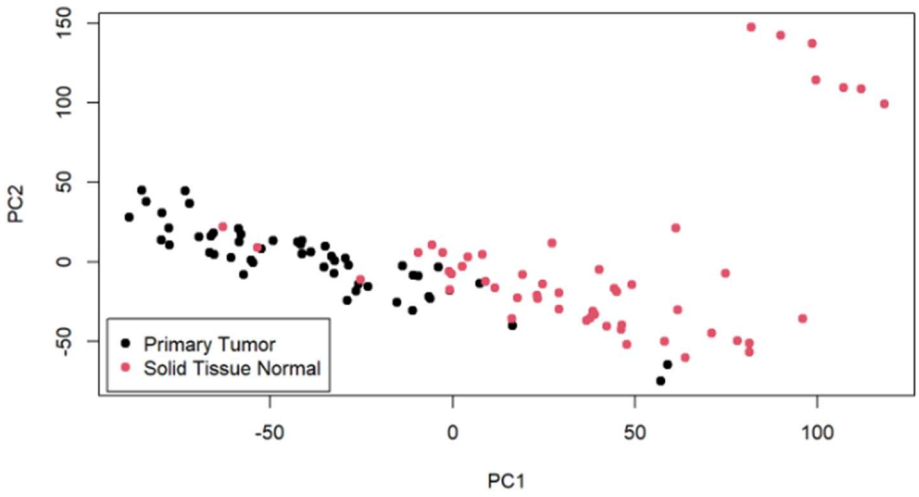


Figure 2:

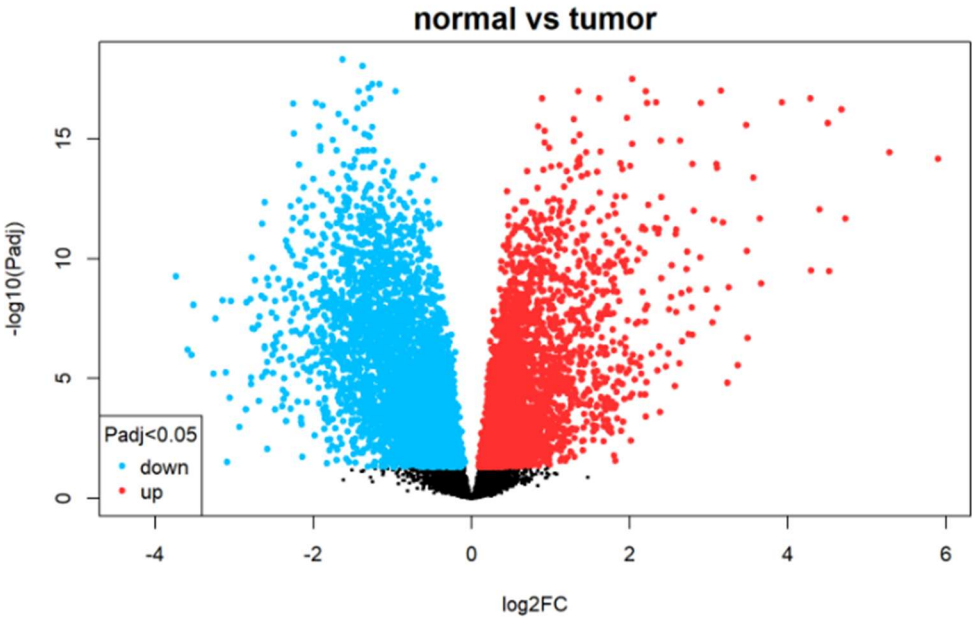


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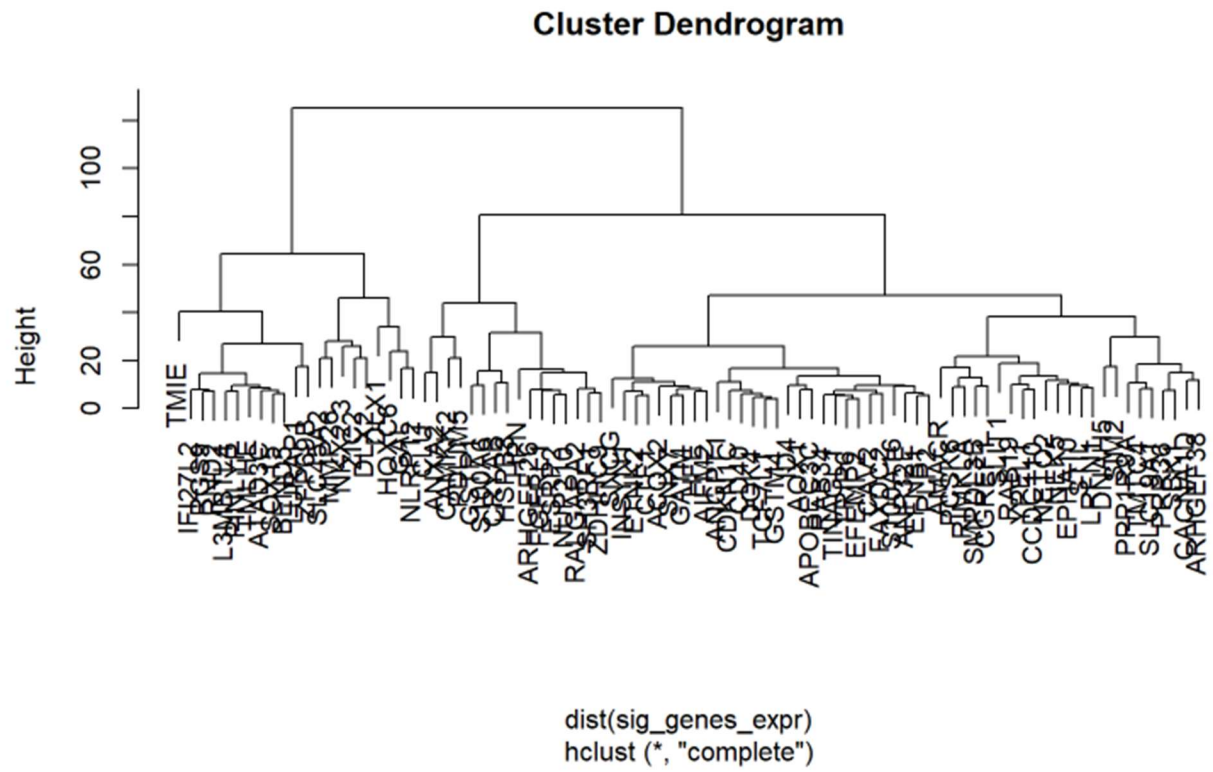
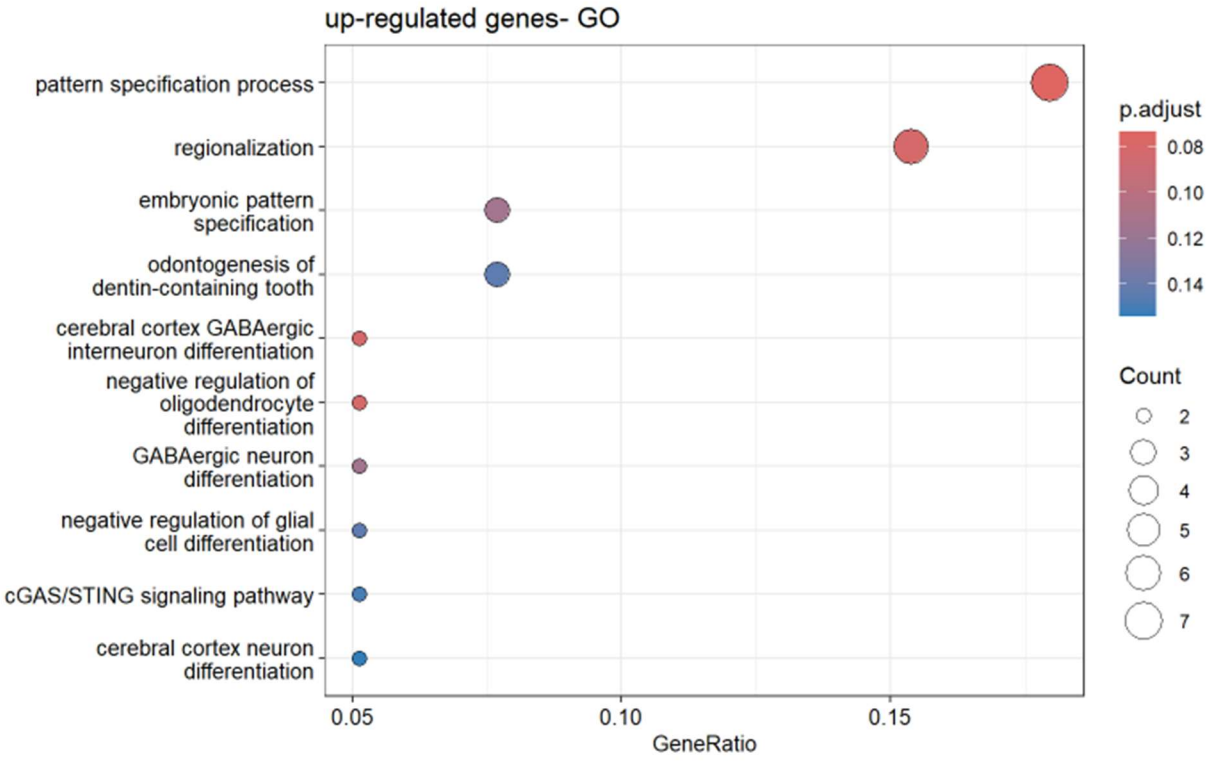


Figure 4 (a – upregulated pathways; b – downregulated pathways)



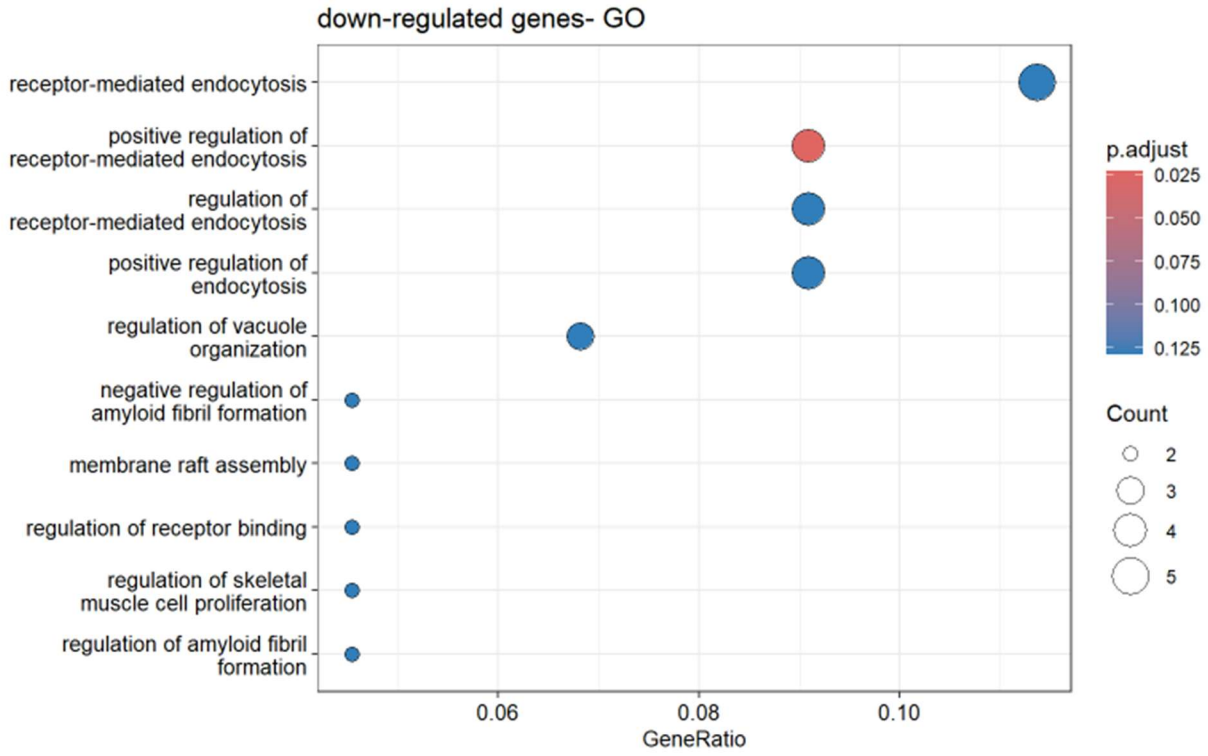
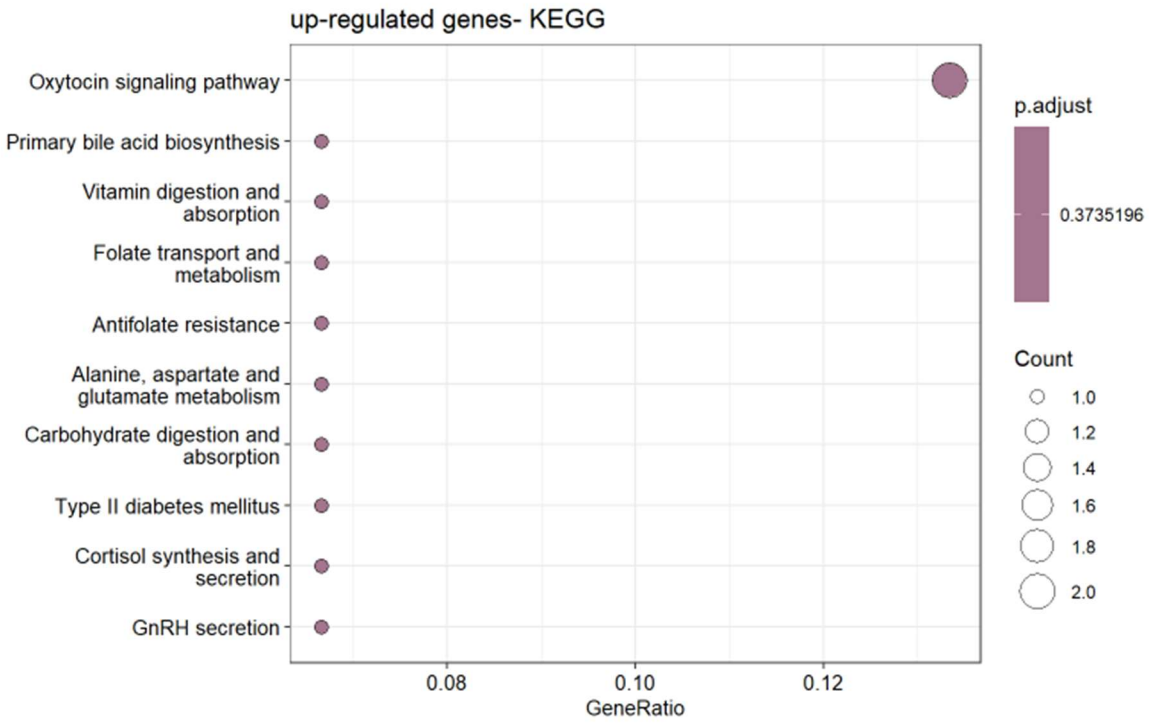


Figure 5 (a – upregulated pathways; b – downregulated pathways)



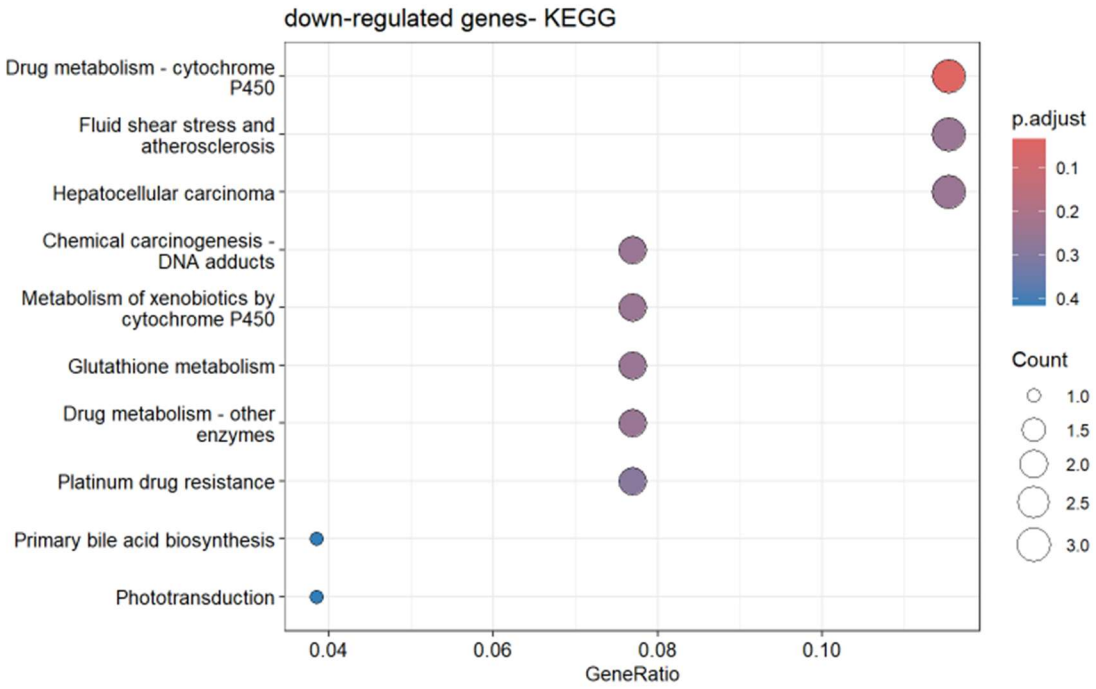
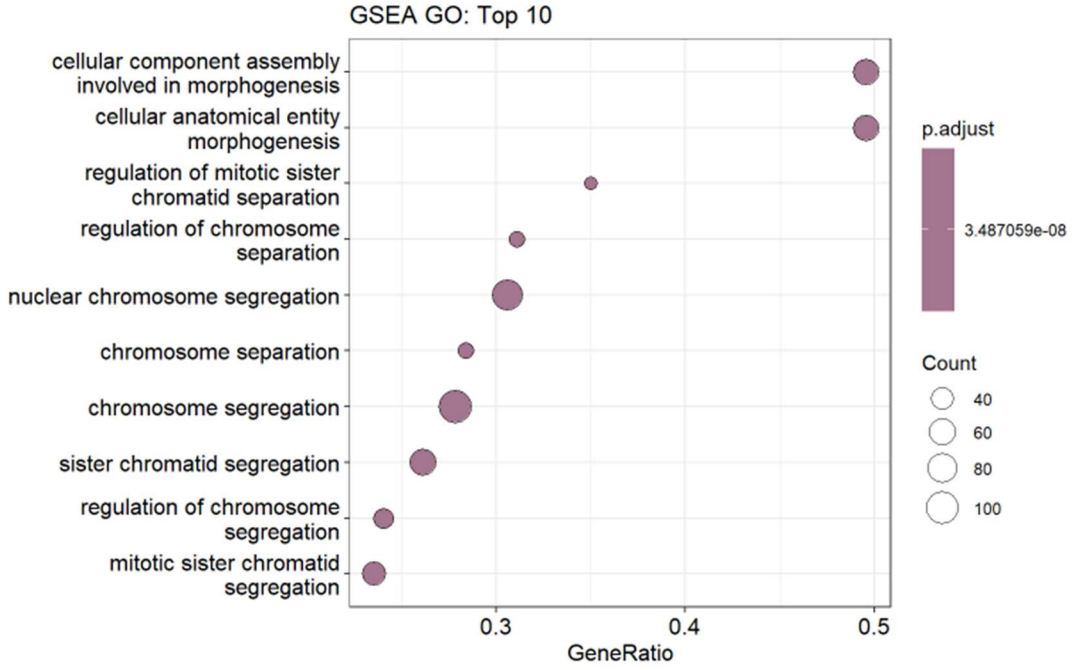

down-regulated genes- KEGG
Drug metabolism - cytochrome P450
Fluid shear stress and atherosclerosis
Hepatocellular carcinoma
Chemical carcinogenesis - DNA adducts
Metabolism of xenobiotics by cytochrome P450
Glutathione metabolism
Drug metabolism - other enzymes
Platinum drug resistance
Primary bile acid biosynthesis
Phototransduction
0.04
0.06
0.08
0.10
GeneRatio
p.adjust
0.1
0.2
0.3
0.4
Count
1.0
1.5
2.0
2.5
3.0

Figure 6) (a –GSEA GO; b – GSEA KEGG)



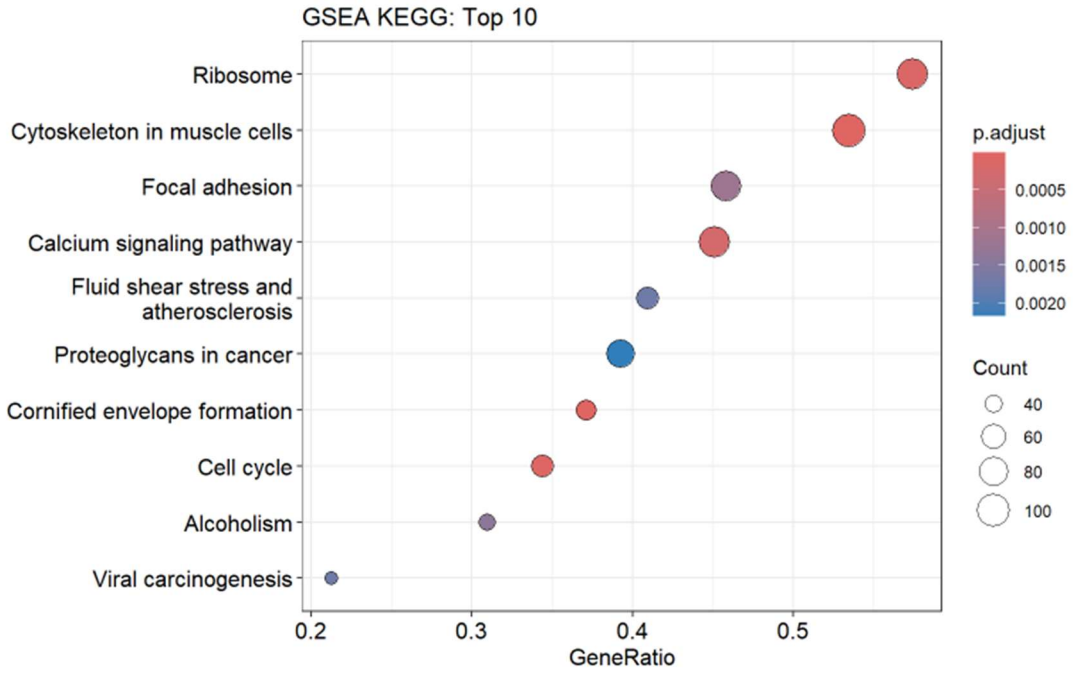


Table 1:

Top 10 incoming hubs

node	in_degree	out_degree	betweenness
APOBEC3C	70	0	0.00000
PINLYP	63	2	11.52552
TMLHE	53	8	20.60891
ZDHHC9	52	4	10.97516
ID4	51	17	38.61318
EPHA10	51	8	55.09689
SBK1	49	6	15.93689
GSTM4	48	24	76.17856
EFEMP2	46	15	50.58938
ANXA2	43	12	28.70983

Top 10 outgoing hubs

node	in_degree	out_degree	betweenness
GSTP1	4	59	15.530638
TRIP6	4	56	11.926720
EFNB1	6	54	15.599312
DOK4	21	53	48.002459
ESRP2	4	51	17.898316
RND2	10	50	15.711416
PLP2	7	49	15.437838
CD40	6	48	11.879524
CRYAB	13	47	18.980028
AIFM2	1	46	1.480985

Figure 7)

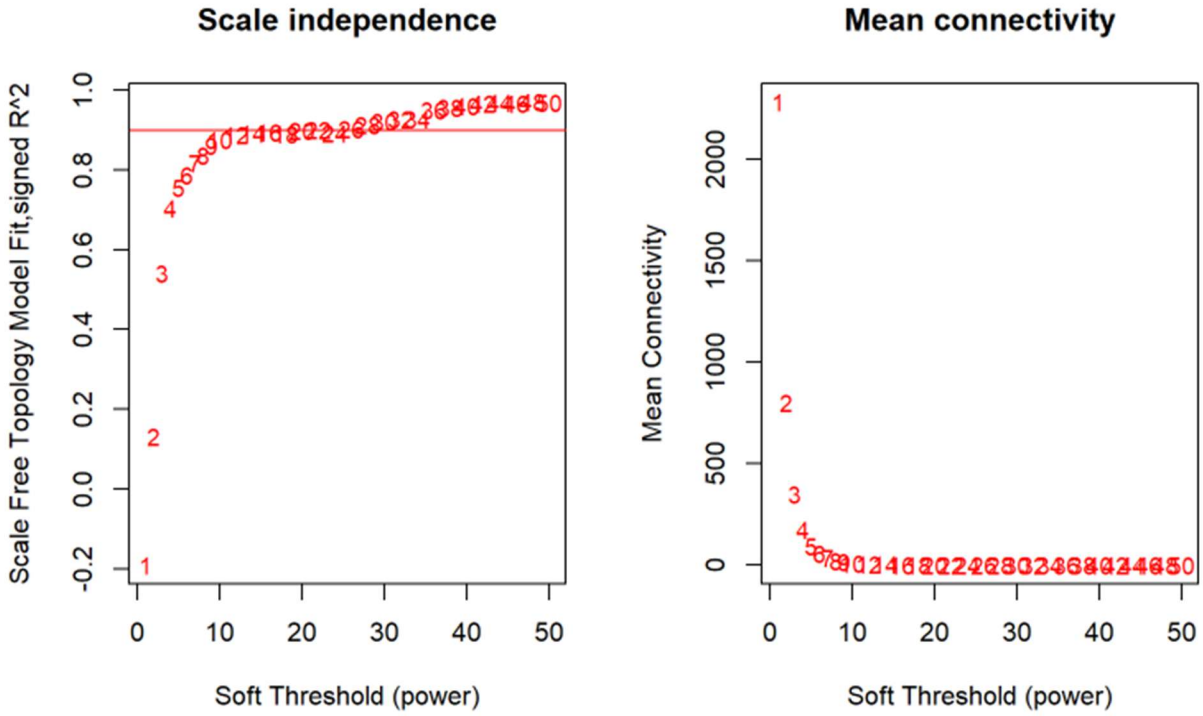


figure 8:

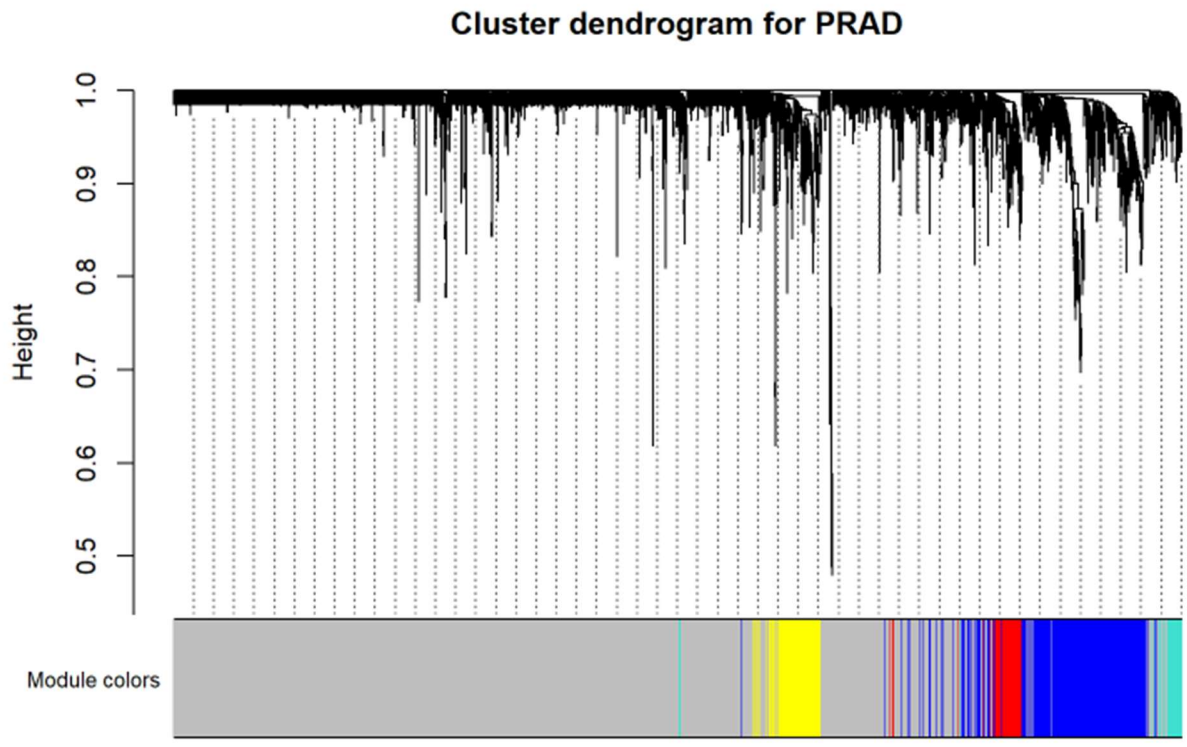


Figure 9

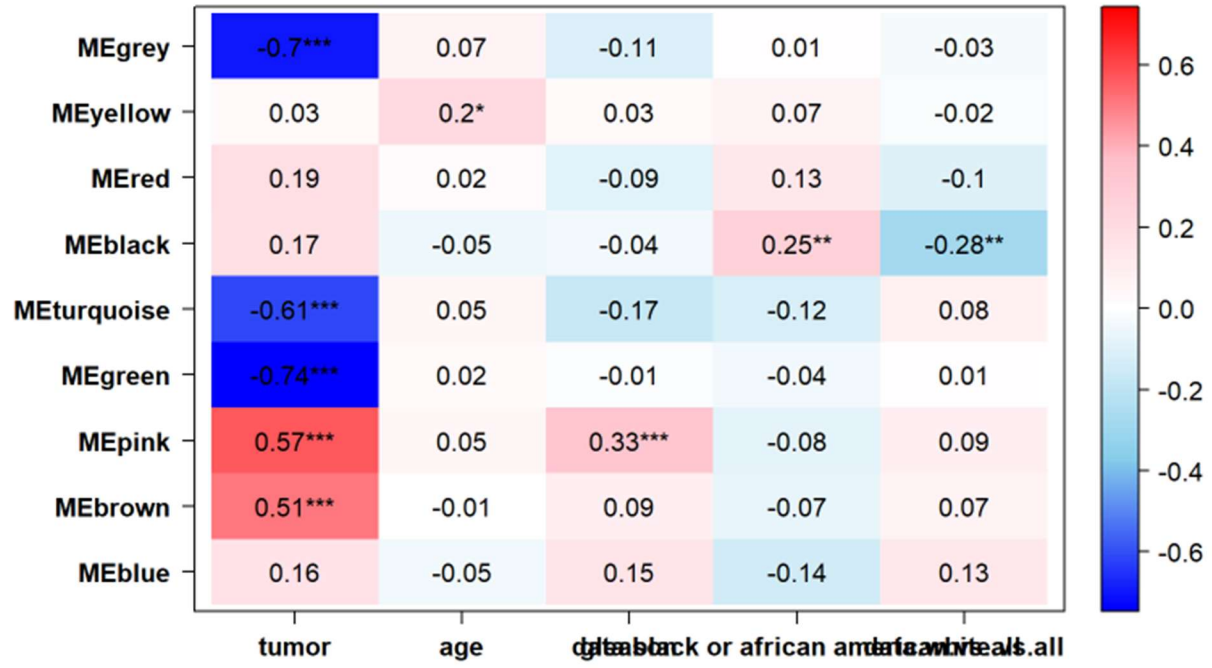
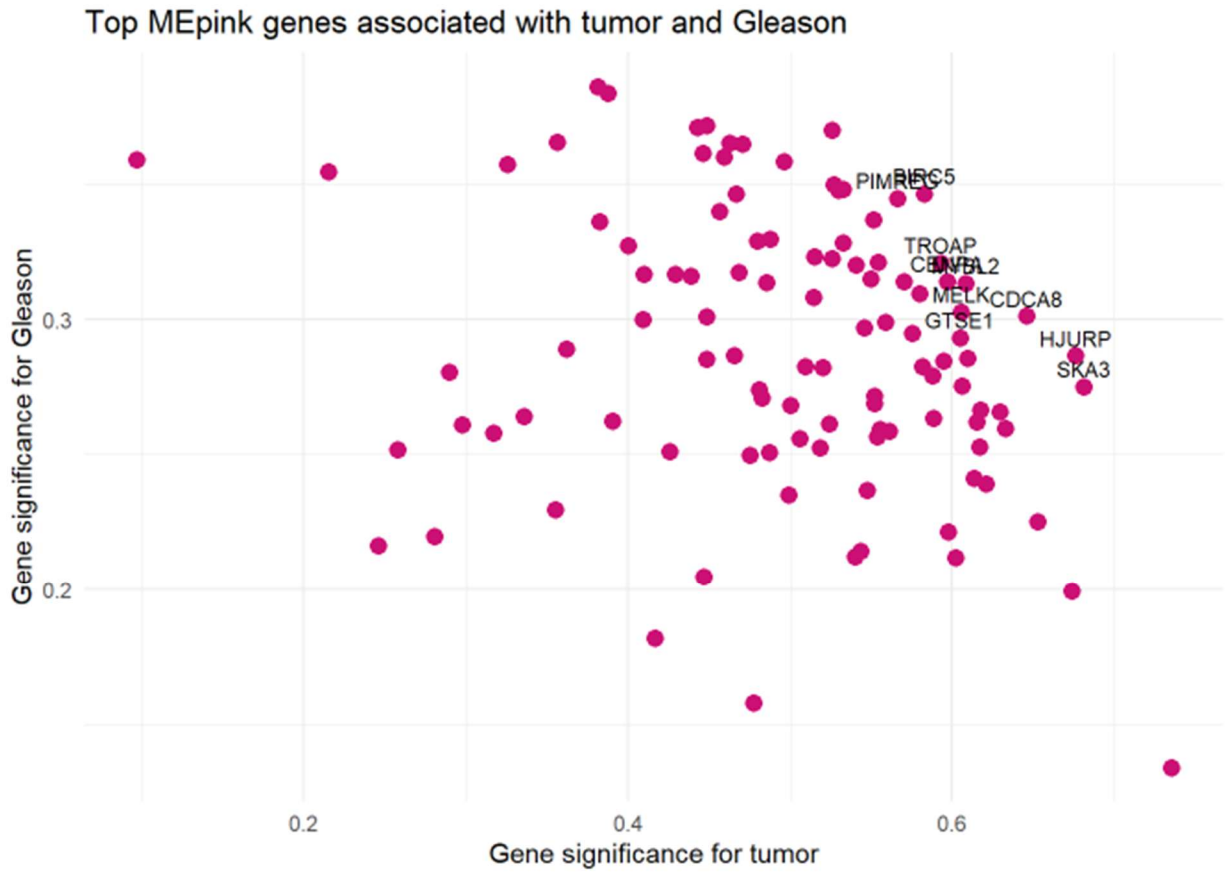

MEgrey
MEyellow
MEred
MEblack
MEturquoise
MEgreen
MEpink
MEbrown
MEblue
-0.7*** 0.07 -0.11 0.01 -0.03
0.03 0.2* 0.03 0.07 -0.02
0.19 0.02 -0.09 0.13 -0.1
0.17 -0.05 -0.04 0.25** -0.28**
-0.61*** 0.05 -0.17 -0.12 0.08
-0.74*** 0.02 -0.01 -0.04 0.01
0.57*** 0.05 0.33*** -0.08 0.09
0.51*** -0.01 0.09 -0.07 0.07
0.16 -0.05 0.15 -0.14 0.13
tumor
age
0.6
0.4
0.2
0.0
-0.2
-0.4
-0.6

Figure 10


Top MEpink genes associated with tumor and Gleason
Gene significance for Gleason
Gene significance for tumor
0.3
0.2
0.2
0.4
0.6
PIMREG
TROAP
MELK
CDCA8
GTSE1
HJURP
SKA3